REVISION INFORMATION

Temperature Control — Trim Air System

Air Conditioning — Trim Air System — Conversion to the Standard 2 Zone Regulation Configuration

Object code	DMC-ZPH-A-21-60-11-00A-700A-A	**Issue**	001-00 (new)
Issue date	2020-02-11	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — In-Service Engineering, Valmont (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	ZEPHYR-20 and ZEPHYR-22 and ZEPHYR-24, all aircraft, post MOD-150422		

Service bulletin number	ZPH-21-1218
ATA chapter	21-60-11
Compliance category	sbcc02
Task category	sbtc03
Original issue date	2020-02-11

REVISION INFORMATION

Revision 01 (2020-02-11): Initial issue

SUMMARY

Conversion of in-service aircraft to the standard 2 trim air regulation configuration, closing the zone loop on a duct-mounted sensor and adding an accumulator to damp the pressure transient that follows a pack change.

1. PLANNING INFORMATION

A. Effectivity

Applicable to ZEPHYR-20, ZEPHYR-22 and ZEPHYR-24 aircraft already at the hardened valve seat standard and not built to the standard 2 configuration. Aircraft delivered from the production cut-in are excluded.

MODEL	SERIAL NUMBERS
ZPH-20C2	ZPH-20C2All aircraft
ZPH-20C5	ZPH-20C5All aircraft
ZPH-20N3	ZPH-20N3All aircraft
ZPH-22C4	ZPH-22C4All aircraft
ZPH-22V2	ZPH-22V2All aircraft
ZPH-22N1	ZPH-22N1All aircraft
ZPH-24C1	ZPH-24C1All aircraft
ZPH-24V1	ZPH-24V1All aircraft

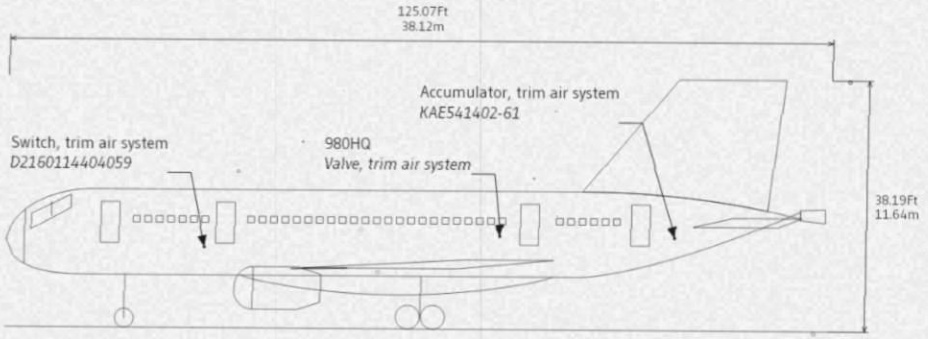
B. Reason

1. PLANNING INFORMATION

Even with the hardened valve seat, cabin zone temperature overshoot on descent averages four degrees and the aft zone of a full aircraft regularly exceeds the selected temperature by more than that. The cause is the position of the sensing element: the loop closes on returned cabin air with a transport delay of two to three minutes, so the valve chases a temperature that is already out of date. Crews on the longest sectors respond by leaving the zone selectors at the extremes, which removes any benefit from the system. Aircraft that are not converted will continue to generate cabin comfort complaints that no valve change can resolve.

C. Description

This Service Bulletin converts the trim air regulation to standard 2. Each zone valve is replaced by the switched assembly that carries a sensor boss on its outlet flange, an accumulator is installed upstream of each valve to damp pack transients, and the zone controller parameter set is re-scaled to the new loop gains. On ZEPHYR-24 aircraft a third sensing point is added for the forward zone. The distribution ducting itself is not replaced.



Temperature Control — Trim Air System — location, side view — Figure 1

D. Compliance

Recommended. Accomplish At the operator's convenience, normally at a base maintenance visit. Operators reporting cabin comfort complaints are advised to embody within 24 months..

E. Approval

The technical content of this service bulletin is approved under the authority of the Aeralis design organisation approval.

F. Manpower

The estimated man-hours are 58 man-hours for one aircraft, based on 3 technicians working at the same time, and an elapsed time of 26 hours. The downtime category is A check.

G. Weight balance

The change of mass is 3.4 kg per aircraft. The aircraft weighing record is to be amended at accomplishment.

2. MATERIAL INFORMATION

H. Interchangeability

The parts are not interchangeable. A mixed installation is not permitted.

I. Publications affected

The following are revised at the next scheduled revision: IPC-Z-21-60-11-FIG-11.

2. MATERIAL INFORMATION

A. Material price

The material is supplied as kit K2160-1218-01, one kit per aircraft.

B. Material kit

Table 1 Kit K2160-1218-01 contents, one kit per aircraft

Part number	Quantity	Unit
D2160114404059	2	EA
KAE541402-61	2	EA
AES510897-5A	8	EA

C. Material availability

The kit is available from 2020-03-16. The following are to be returned: KAD437362-51. Consumables: Grease, high temperature, silicone free, Sealant, polysulphide, class B. Tooling: Torque wrench 3-20 Nm, Portable data loader. Kit K2160-1218-01 contains two switched assemblies, two accumulators, the bracket and pin set and the sensor harness for a two-zone aircraft; kit K2160-1218-02 adds the third channel for ZEPHYR-24 aircraft. The switched assembly is a long-lead item at twenty-eight weeks. Removed valves are returned to Aeralis for credit.

3. ACCOMPLISHMENT INSTRUCTIONS

A. General

Conversion of in-service aircraft to the standard 2 trim air regulation configuration, closing the zone loop on a duct-mounted sensor and adding an accumulator to damp the pressure transient that follows a pack change.

B. Procedure

1. Confirm that the aircraft is at the hardened valve seat standard before you start; if it is not, this bulletin must not be embodied.

2. Make sure that the air conditioning system is depressurised.

3. Get access through the ceiling and lateral panels in zones 254, 291 and 273.

4. Remove the existing valves following task AMM-Z-21-60-11-000-023-A.

5. Install the switched assembly D2160114404059 at the forward valve FIN 726HQ and at the aft valve FIN 980HQ.

6. On ZEPHYR-24 aircraft install the third assembly at FIN 872HQ.

7. Install the accumulator KAE541402-61 on the new bracket and secure it with the clevis pins AES510897-5A.

4. ADDITIONAL INFORMATION

8. Install the valves following task AMM-Z-21-60-11-400-023-A.

9. Connect the sensor harness at the duct boss and do a continuity check.

10. Load the revised zone controller parameter set.

11. Do the trim air system operational test with the packs running and record the zone response time.

12. Restore the ceiling and lateral panels.

4. ADDITIONAL INFORMATION

A. Para

Report the accomplishment of this service bulletin through the customer services portal, quoting the aircraft serial number and the date of accomplishment.